

Apache Kafka

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- Developed by LinkedIn and kafka.

④ Problem Solved

→ In real world application especially where Real Time Communication is very high, direct I/O to Database becomes a Bottleneck (Because of Low Throughput of DBs)

④ Throughput

→ UOPS (Operation per second)

For example, (Pathao)

At any time, there will n pathao users i.e. (n drivers and n users)

$$[0 < n < 1000]$$

In such case, the users need the real time location of the Drivers.

If there are 500 drivers, each driver needs to send its [location, speed and other] to the Database so that it can be used for Analytics, Real Time Update to the user.

Per second 500 data needs to be written in the database, and DBs don't have high throughput. This is not scalable.

So, we need a service with very High Throughput. So, that we can later bulk insert all those data in a single transaction.

- (A) Apache kafka
- service with High Throughput
 - But less Storage [Temporary]

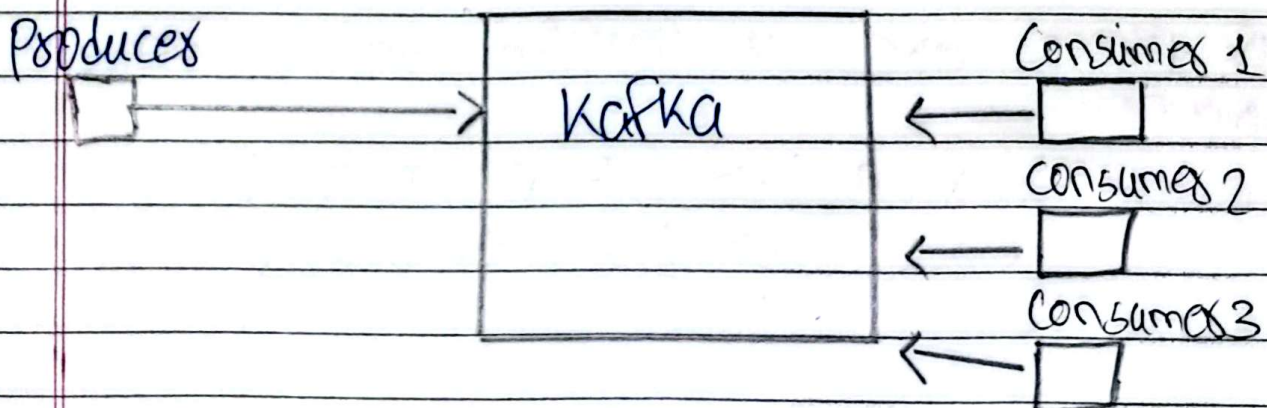
On the other hand,

DBs have Long term storage with less throughput.

- (A) Solution
- Combine Both kafka and DB.

So, all the real time data will be dumped into the kafka. Each second or periodically we'll bulk insert the collected data in a single Transaction.

(A) Architecture



We can partition the data produced by Producers in Kafka using Topics.

For e.g:

Location $\Rightarrow$ Topic

Speed $\Rightarrow$ Topic

We can further divide the Topic data using Partition.

- Also we can have, Consumers and Producers for each partition as well.
- Each [Partition] has a unique [Index] to identify it
- [Producers] will already have idea about which ~~Partition~~ [Partition] to use for data dumping.

⑧ Consumers Groups

- We can create Groups for consumers. These can be multiple @ Group.
- Uses Auto Balancing to balance the Partition access to each Group.

If there are 4 Partition and two groups.

Group A $\Rightarrow$ [5 members]

Group B $\Rightarrow$ [3 members]

For G1,

- Each partition will be provided to a single consumer so, extra will not have a partition.

For G2,

- As there is 4 partition and, 3 members any 2 partition will be assigned to a single member and single to each one of two members.

④ Queue vs Pub/Sub

- Queue
1 Producer, 1 Consumer
e.g. RMA,
- Pub/Sub
Many Producers, Many consumers
- Kafka is Both Queue and Pub/Sub.
We can use any. •